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THE ULTIMATE GUIDE TO

Saving Data Egress Costs in the Cloud

Introduction

Cloud storage has gained popularity in recent years, but managing costs can be a significant challenge. While major public cloud providers, such as AWS, Google Cloud, and Microsoft Azure, offer free data input to the cloud, they impose high fees for data retrieval from the cloud, also known as data egress fees.

Data egress fees are retroactively charged and are therefore often considered hidden fees. This means that your applications, workloads, and end-users may unknowingly accumulate high fees until you receive the bills. For instance, if your data platform architecture spans across cloud regions, hybrid cloud, or multi-cloud, egress fees will apply.

In fact, egress costs have affected 34% of enterprises using cloud storage[1]. Managing data egress fees is challenging, particularly in large enterprises where multiple offices, departments, or branches conduct data analytics.

This guide explains what you need to know about data egress costs and five proven best practices for saving your egress costs.

[1] S&P Global Market Intelligence, Data center interconnection faces cloud-native competition, March 15 2022.

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Chapter 1.
What are Data Egress Fees?

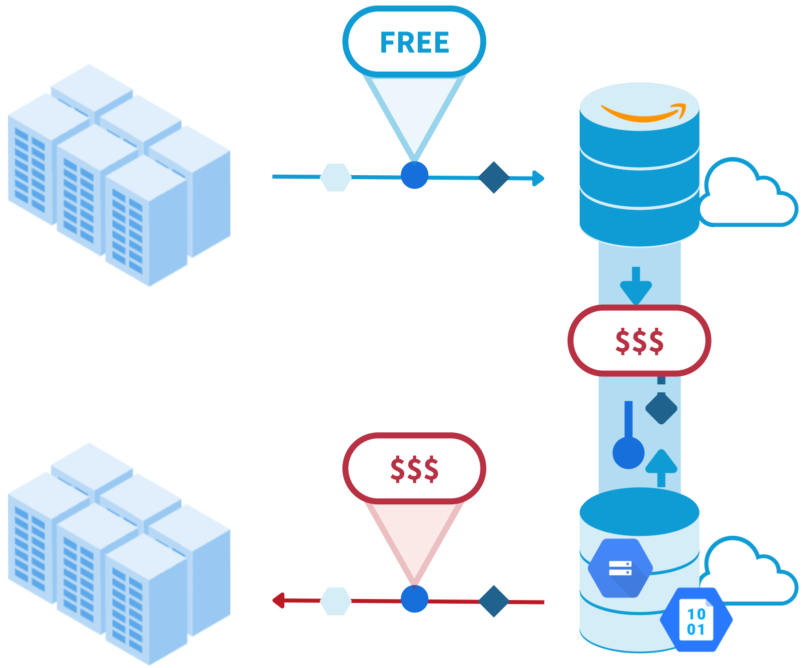
1.

Cloud Data Egress Terminology

Before delving into ways to reduce egress costs, let's define several key terms.

- **Data transfer** is a generic term that refers to any type of movement of data over the network. The movement can be within the same cloud or between a cloud and an external location, such as another cloud or on-premise infrastructure. Data transfers involve moving data into the cloud or out of the cloud.
- **Ingress**, or inbound, refers to the transfer of data into a cloud.
- **Egress**, also known as outbound, refers to the transfer of data from a public cloud to another type of infrastructure, typically an on-premises datacenter.
- **Bandwidth**, in the context of cloud data storage costs, is a general term used to describe the transfer of data from one location to another. While in other contexts, bandwidth can refer to the throughput of a network, it typically refers to data transfer when discussing terms like "Azure bandwidth pricing."

Note that AWS refers to Egress/Ingress as Data Transfer Out/In, respectively, while Google Cloud and Microsoft Azure use the terms Egress/Ingress.



2.

How Do Cloud Service Providers Charge Data Egress Fees?

We examined the pricing and conditions of storage services offered by the leading cloud providers, Amazon S3, Google Cloud Storage, and Azure Blob Storage.

As shown in the table below, transferring data into the clouds (ingress) is free for all three providers. In contrast, the data egress fees for pulling data out of clouds are calculated based on the volume and vary based on the pricing terms.

Cloud Service Provider	Ingress	Egress	
		Traffic (per month)	Pricing
Amazon S3*	Free	100 GB - 10 TB	$0.09/GB
		10 - 50 TB	$0.085/GB
		50-150 TB	$0.07/GB
		> 150 TB	$0.05/GB
		> 500 TB	Contact sales
Google Cloud Storage**	Free	0 - 1 TB	$0.12/GB
		1 - 10 TB	$0.11/GB
		> 10 TB	$0.08/GB
		Customized	Contact sales
Azure Blob Storage***	Free	100 GB - 10 TB	$0.087/G
		10 - 50 TB	$0.083/GB
		50-150 TB	$0.07/GB
		150 - 500 TB	$0.05/GB
		> 500 TB	Contact sales

Pricing applies to AWS US East Ohio region, GCP North America regions, Azure North America regions. As of March 9, 2023.

*: https://aws.amazon.com/s3/pricing/

**: https://cloud.google.com/storage/pricing#network-egress

***: https://azure.microsoft.com/en-us/pricing/details/bandwidth/#pricing. Azure charges more egress fees if you are transferring data from Asia. Also, the fees are lower if you use a preferred ISP for the transfer.

However, there are instances where no egress fees are charged. Generally, data transfers within the same cloud region are free, with Azure being more restrictive, limited to transfers within Availability Zones. Additionally, the initial 100GB of data transfer per month is usually free.

Cloud Service Provider	No Egress Fees When
Amazon S3*	• Data transferred out to the internet for the first 100GB per month, aggregated across all AWS Services and Regions (except China and GovCloud) • Data transferred between S3 buckets in the same AWS Region • Data transferred from an Amazon S3 bucket to any AWS service(s) within the same AWS Region as the S3 bucket (including to a different account in the same AWS Region) • Data transferred out to Amazon CloudFront
Google Cloud Storage**	• Data moves within the same location • Data moves from a Cloud Storage bucket located in a dual-region to a different Google Cloud service located in one of the regions that make up the dual-region • Data moves from a Cloud Storage bucket located in a region to a different Google Cloud service located in a multi-region, and both locations are on the same continent • 100 GB per month from North America to each Google Cloud egress destination (Australia and China excluded), aggregated across US-WEST1, US-CENTRAL1, and US-EAST1 regions
Azure blob Storage***	• Data transfer within same Availability Zone (Starting from July 1, 2023, Data transfer billing between Virtual machines across availability zones will begin) • Data transfer from Azure origin to Azure CDN • Data transfer from Azure origin to Azure Front Door • First 100GB / month: from North America, Europe to any destination; from Asia (China excluded), Australia, MEA to any destination; from South America to any destination

Pricing applies to AWS US East Ohio region, GCP North America regions, Azure North America regions. As of March 9, 2023.
*: https://aws.amazon.com/s3/pricing/
**: https://cloud.google.com/storage/pricing#network-egress
***: https://azure.microsoft.com/en-us/pricing/details/bandwidth/#pricing. Azure charges more egress fees if you are transferring data from Asia. Also, the fees are lower if you use a preferred ISP for the transfer.

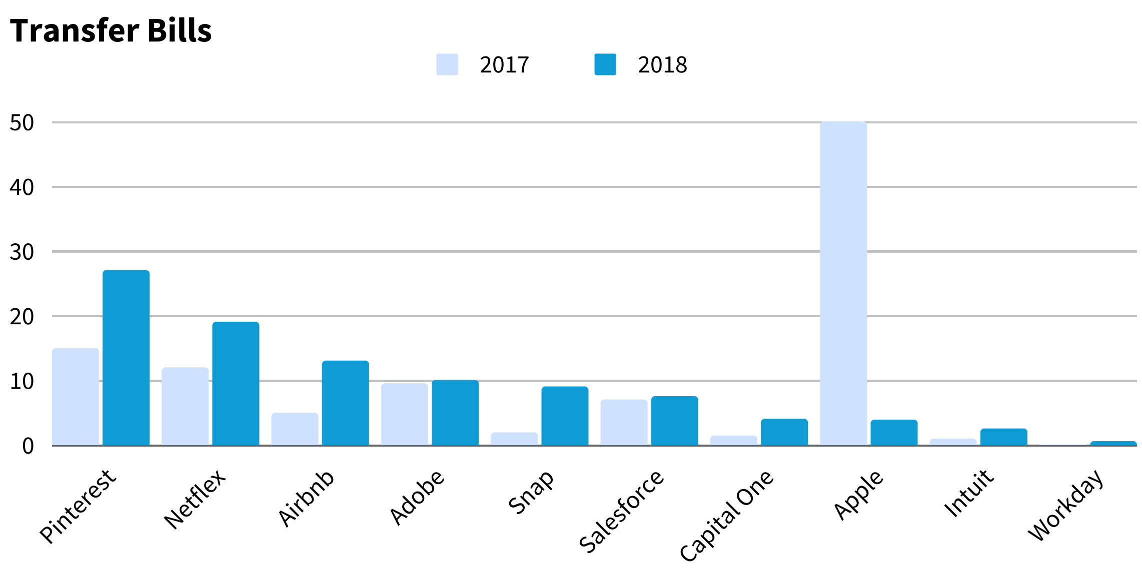
3.

Controlling Data Egress Cost is a Common Challenge

As data transfers occur frequently, so do egress fees. Managing the cost of data egress is a common challenge faced by organizations.

An IDC survey[2] revealed that almost everyone (99%) incur planned or unplanned egress fees, and 41% incur these fees frequently. The impact of these charges is further highlighted by a study conducted by S&P Global[3], which found that 34% of enterprises have had to repatriate data on-premises or switch to a provider that doesn't charge for egress.

The bills are significant for large enterprises. Apple spent $50 million in data egress fees in one year, while Pinterest spent over $20 million, according to The Information[4]. Netflix and Airbnb also incurred data transfer charges exceeding $15 million. Additionally, Adobe, Snap, and Salesforce paid more than $7 million each.



[2]IDC, Future-proofing Storage: Modernizing Infrastructure for Data Growth Across Hybrid, Edge, and Cloud Ecosystem, March 2021.
[3]S&P Global Market Intelligence, Data center interconnection faces cloud-native competition, March 2022.
[4]The Information, AWS Customers Rack Up Hefty Bills for Moving Data, October 2019.

Chapter 2.
Best Practices for Saving Your Egress Cost

To control the costs of data egress, cloud and data infrastructure professionals have tested and proven the effectiveness of the following five best practices.

1.
Use Data Caching to Avoid Unnecessary Roundtrips

Data caching is a very effective technique for egress cost reduction because it reduces data round trips across the network.

Caching offers several benefits. First, it reduces network traffic associated with data transfer between different regions, data centers, or clouds. Second, it improves application performance, as data is available locally and can be accessed performantly without having to wait for remote access. Thirdly, caching can help to reduce load on the storage side by decreasing the number of requests made to it.

Caching works by bringing data closer to applications. You can cache frequently accessed data in a closer and faster storage layer such as solid-state drives (SSD). When a data request is made, the cache is checked first to see if the data is present. If it is, the data is immediately returned without needing to be fetched from the original under storage. Otherwise, it is fetched from the original source and cached for future use.

An access-based caching strategy is recommended to keep data movement on-demand. This allows caching to make decisions on what data to store and how long to store it instead of manually identifying the exact data that needs to be accessed. Access-based caching can help improve performance and reduce latency by storing frequently accessed data in an optimal storage tier that is close to the application.

An estimation shows that using access-based caching can result in up to an 80% reduction in egress costs in a hybrid cloud environment (data resides in the GCP while compute is on-premises) compared to directly accessing data.

PRO TIP ► By using data caching, Expedia has reduced cross-region data transfer costs on Amazon S3 by 50%. They stopped cross-region replication of petabytes of data per month. Read the full story>>

2.

Streamline Your Data Pipeline to Minimize Data Replications

When designing your data pipeline, it's essential to consider efficient ways to handle data flow, minimize data duplication and streamline your pipeline associated with egress to save costs.

Efficiency should be a guiding principle in your data architecture, and one of the goals should be to avoid unnecessary data copies, duplication of components, or network traffic. Be cautious with data replication in your data pipeline, as copying data across geo-distributed storage systems generates data traffic that is subject to data egress fees. In addition, data copies in pipelines are not only redundant, but also complex as adapting these pipelines for constant updates to data in the cloud is challenging.

For data replication across regions or clouds, it is advisable to replicate only the changes in the data, known as deltas. To accomplish this, you can consider implementing data management policies that are based on the date of creation or only capture updated data. This policy-driven approach helps perform incremental data synchronization, resulting in lower egress costs as the amount of data transfer is minimized.

To keep data flow to a minimum, it's also advisable to use data compression techniques to reduce the size of data being transferred. By compressing data, it's possible to reduce the amount of data that needs to be transferred, thereby reducing egress costs.

PRO TIP ► Best practices include avoiding cross-region data replication and use policy-based data management across geo-distributed storage.

3.

Optimize Data Outflow of Your Architecture

It is essential to understand and optimize the data outflow points in your architecture to minimize data transfer charges. Egress fees arise primarily when resources are dispersed across multiple regions or availability zones. Also, the charges vary across different regions and zones.

Keep in mind that egress costs increase as the destination moves further away. The most expensive rates apply to data transfers from the cloud to the internet. There are almost always charges for data transfers between cloud regions, and the cost of transfers between availability zones within a region depends on the cloud service being used. There are no costs associated with data transfers within a single availability zone, as mentioned in the previous section.

To reduce data transfer costs, design an infrastructure that follows the least expensive routes. Try to minimize traffic to the internet and between regions and availability zones while maximizing traffic within an availability zone, or at the very least, within a region. This is especially crucial when decoupling compute resources and storage for flexibility.

Unless mandated by compliance or security requirements, select the geographic region with the lowest egress pricing. Typically, regions within the United States and Canada have the most affordable rates, whereas Singapore, India, and South America are among the most expensive. Unless there is a compelling business reason to host your services in these countries, consider opting for the more cost-effective North American regions.

PRO TIP ► The further away the destination, the more expensive egress costs are. Keep data transfers within the same region in your data pipeline whenever possible.

4.

Monitor Your Egress Fees Regularly

Regularly monitoring egress fees is crucial to prevent unexpected bills and minimize disruptions to your cloud billing. Having observability helps you understand where your egress costs are allocated and take appropriate action before they become a significant financial burden.

One effective way to achieve this is by adding cost allocation tags to your instances and load balancers. AWS provides a feature called cost allocation tags, which can be used in conjunction with the AWS Cost Explorer to analyze data transfer fees stemming from instances and services. By tagging, you can narrow down which ones generate the highest AWS data transfer costs and further filter them by type: CloudFront (out), Inter AZ, Internet (In), Internet (Out), Region to Region (In), and Region to Region (Out).

After filtering your data transfer costs, it's important to observe the trend over multiple months to identify any sudden increases or spikes in costs. Grouping your data by the Name tag can help you identify which instances or services contribute to the increase in your data transfer costs.

PRO TIP ► Understanding where your egress costs are allocated and regularly monitoring them is a proactive approach to cloud billing management that can help you save costs in the long run.

5.

Negotiate Customized Pricing with Your Cloud Provider

If you require data transfers exceeding 500 TB per month, it's recommended to negotiate a customized quote for organization-specific deals.

Private Pricing Programs, like AWS's, allow you to commit to using a minimum amount of data transfer each month between designated regions for services. For instance, if you anticipate requiring one Petabyte's worth of monthly data transfer over the next year and have optimized as much as possible, a Private Pricing Program can secure a discounted overall cost in exchange for committing to that minimum usage amount.

Note that failing to utilize the full amount committed still obligates you to pay for it. However, committing to a minimum usage amount can ensure predictable egress costs and lower prices, especially if your data transfer needs are clear over a specific period.

PRO TIP ► A large Fintech company negotiated a custom quote of $0.045/GB egress fees per month with Google Cloud.

Conclusion

To summarize, we have compiled a list of best practices to aid in reducing egress costs. These practices include implementing data caching, improving pipeline design, limiting bandwidth, monitoring, and negotiating. It is important to use them wisely based on your business needs to avoid unexpected data transfer charges.

Controlling egress costs is important, but it's only one aspect of part of cloud cost optimization. More importantly, ensure that your spending on all cloud services corresponds to and promotes your business objectives and strategies. Furthermore, remember that cost optimization is a continual process because the cloud is continuously developing. Therefore, remain watchful and continue to optimize your expenses.

GET A COST ANALYSIS ► Discover how you can reduce your cloud costs by up to 80%. Get your analysis today >>

About the Author

Hope Wang has a decade of experience in Data, AI, and Cloud. She is an AWS Certified Solutions Architect – Professional. She is also an open-source contributor to Alluxio, Trino, and PrestoDB. She earned a BS in Computer Science, a BA in Economics, and an MEng in Software Engineering from Peking University, as well as an MBA from USC.

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